

## Conditioning tensors

Historical PM2.5

$X_t: B \times 168$

Origin context

$q_t: B \times 85$

Retrieval summary

$\rho(G_t): B \times 51$

Drift components + score

$d_t: B \times 6$

Future calendar

$c_{t+h}: B \times 9$

## Forecast paths

24 direct XGBoost  
regressors (M09)

$B \times 151 \rightarrow B \times 24$

Frozen Chronos-Bolt

$B \times 168 \rightarrow B \times 24$

Drift diagnostics  
score + regime flag

$D_t \in \{0,1\}$

Forecast-level fusion

$\omega=0.75$

$\hat{Y}_t$

Two verified forecast variants with diagnostic drift evidence

## Evidence output

Signed XGBoost effects

top 3 fields

Retrieved window IDs

top 3 cases

Recent event IDs

prior 72 h

Drift + uncertainty proxy

machine-readable

7,199 evidence records